Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

Team name: Myanmar Dev

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Problem Statement: Hybrid and distributed organizations across JAPAC face severe recognition fragmentation: critical employee milestones (birthdays, work jubilees) are siloed in manual spreadsheets, standard SaaS tools lack native localized greetings across diverse Asian cultures, and HR administrative friction leads to delayed or forgotten celebrations and reduced workforce engagement.

Brief about the idea

Enterprise AI Milestone & Culture Platform

• Zero-Touch Automation: Serverless Cloud Functions daily cron at 09:00 AM (Asia/Rangoon) automatically evaluates workforce records for birthdays and tenure jubilees (>= 1 year).

• Culture-Aware Gemini AI: Google Gemini (gemini-2.5-flash) synthesizes authentic, culturally attuned celebratory greetings tailored to each employee's language and tenure.

• Omnichannel Delivery Hub: Zero-touch multi-channel dispatch via SMTP Email, Microsoft Teams Adaptive Cards, and Telegram Bots.

• 3D Interactive Network: Real-time Three.js WebGL globe mapping 34 JAPAC territory nodes with Great-Circle flight arcs connecting regional offices to the Singapore hub.

Built Specifically for Asia-Pacific Teams

• 7 Native JAPAC Locales: 100% translation parity across English, Burmese, Vietnamese, Traditional Chinese, Cantonese, Korean, and Japanese with script-safe typography.

• Client-Side Slip OCR: On-device computer vision receipt scanner supporting regional mobile wallets (KBZ Pay, Wave Pay, AYA Pay, CB Pay) + multi-currency checkout.

• Lean Performance Architecture: Rollup chunk splitting and route-level lazy loading reducing entry bundle from 1,820 kB to 366 kB (112 kB gzip).

• 1-Click Judge Demo Mode: Instant evaluation mode with local storage persistence and mock telemetry requiring zero cloud credential setup.

Opportunities & Market Differentiation

How Different from Existing Ideas?

✔ Legacy HR Limitation: Workday, Lattice, and BambooHR enforce US/EU corporate workflows with rigid English-only templates and weeks of onboarding.

✔ **Regional Native Parity:** Milestone is engineered ground-up for JAPAC with 7 native regional languages and cultural celebration etiquette.

✔ **Zero-Config Ingestion:** Instant SheetJS Excel/CSV parsing with automated column detection and Metabase-style external database connectors.

How Will It Solve the Problem?

✔ 100% Zero-Touch: Eliminates administrative reminder friction via autonomous daily Cloud Functions cron and Gemini AI copy generation.

✔ **Cultural Relevance:** Celebration copy is customized to employee tenure, role, and local culture, eliminating awkward machine-translated greetings.

✔ **Omnichannel Presence:** Delivers greetings straight to where distributed teams collaborate (Email, MS Teams, Telegram).

Unique Selling Proposition (USP)

✔ Contextual Gemini AI: Generative celebration engine honoring local Asian holidays, tenure milestones, and linguistic nuances.

✔ **3D WebGL Workforce Globe:** Interactive 3D spatial visualization of cross-border team connectivity and milestone density.

✔ **Client-Side Wallet OCR:** In-browser computer vision verification for local mobile wallet payment slips at $0 cloud API cost.

List of features offered by the solution

1. Automated Milestone Detection & Cron

Daily Cloud Functions cron at 09:00 AM (Asia/Rangoon) continuously calculates birthdays and completed work anniversaries (>= 1 year) across all registered tenants.

2. Google Gemini Contextual Copywriter

Gemini 2.5 Flash crafts warm, hyper-personalized celebration messages adapted to employee tenure, seniority, and preferred JAPAC language.

3. 3D Interactive JAPAC Earth Network

Three.js WebGL globe visualizing 34 regional territories with camera-anchored lighting and elevated Great-Circle flight arcs connecting regional nodes.

4. Omnichannel Delivery Engine

Multi-channel dispatching hub supporting SMTP Email (Gmail, custom host), Microsoft Teams Adaptive Cards, and Telegram Bots.

5. Smart Roster & Database Ingestion

Instant SheetJS Excel/CSV parsing with automated column mapping, inline validation, and Metabase-style external database connectors.

6. Client-Side OCR & Local Wallet Checkout

Multi-currency plan upgrades with on-device computer vision OCR scanning KBZ Pay, Wave Pay, AYA Pay, and CB Pay transaction slips.

7. Executive Analytics & Velocity Radar

Interactive Recharts dashboard displaying 7-day recognition velocity, team milestone radar, and cross-border headcount distribution.

8. Enterprise RBAC & Multi-Tenant Quotas

4 role tiers (user, pro, pro_plus, admin) with Firestore security rules, tenant employee limits, and superuser broadcast messaging.

Process flow diagram or Use-case diagram

Primary Flow: Autonomous Daily Milestone Recognition Lifecycle

1. Data Ingestion
HR syncs roster via Excel/CSV or external DB; records encrypted in Firestore.

2. Daily Cron
Cloud Scheduler invokes Cloud Function at 09:00 AM (Asia/Rangoon).

3. Milestone Engine
Calculates birthdays and anniversary jubilees (>= 1 year).

4. Gemini AI Copy
Gemini 2.5 Flash crafts personalized greetings in native language.

5. Omnichannel Dispatch
Sends via SMTP Email, MS Teams Webhook, and Telegram Bot.

6. Telemetry Log
Delivery status logged in Firestore; updates Executive Dashboard.

Secondary Flow: Multi-Currency Plan Upgrade & Client-Side Payment Slip OCR

A. Plan Selection
User selects Pro/Ultra tier; picks regional currency (USD, MMK, SGD, THB).

B. Dynamic QR Payment
System generates local wallet QR code (KBZ Pay, Wave Pay, AYA Pay) with 5-min timer.

C. Client-Side Slip OCR
HTML5 Canvas computer vision verifies luminance, text edges, amount & transaction ID.

D. Instant Activation
Zero cloud API fees; instant quota upgrade applied to tenant account in Firestore.

Wireframes/Mock diagrams of the proposed solution

1. 50/50 Cinematic Hero Split (Login)

- Visual Earth Viewport: Full-height WebGL Earth orienting to Singapore/JAPAC sunrise shader with territory nodes.
- Glassmorphic Auth Card: Google OAuth, passwordless magic links, and 1-Click Judge Demo Mode button.
- Interactive Language Bar: Visible pill navigation to switch across 7 regional languages instantly.

2. Executive Recognition Dashboard

- KPI Metric Tiles: Real-time employee headcount, upcoming 7-day milestones, and delivery success rate.
- Recognition Velocity: Recharts multi-series area chart tracking weekly recognition throughput.
- Upcoming Radar: Chronological milestone feed with 1-click manual greeting preview and trigger.

3. Employee Roster Management Hub

- Tabular Grid: Editable employee roster with instant toggle switches for automated milestone emails.
- SheetJS Import: Drag-and-drop Excel/CSV importer with automated column header detection and preview.
- Metabase DB Connector: Wizard connecting external PostgreSQL, MySQL, and BigQuery HR databases.

4. Omnichannel Settings & Slip OCR

- Channel Config: Dedicated tabs for Email SMTP (Gmail/Custom), MS Teams Webhooks, and Telegram Bots.
- Live AI Preview: Real-time Gemini greeting card preview rendered in the employee's native language.
- Fintech Checkout: Dynamic payment QR code with on-device canvas OCR receipt scanning.

Architecture diagram of the proposed solution

Layer 1: Presentation & Client Layer (React 19 + Three.js WebGL)

Core UI Stack: React 19.0, TypeScript 5.8, Vite 6.4, Tailwind CSS v4, Lucide React icons. | 3D JAPAC Globe: Three.js r160 WebGL Earth with GeoJSON border mapping, front-facing lighting, and SLERP arcs. | Enterprise UX: Recharts 2.15 analytics, i18next supporting 7 native JAPAC locales with script-safe typography.

Layer 2: Intelligence & Client-Side Processing Layer

Generative AI: Google Gemini (gemini-2.5-flash) synthesizing personalized, culture-aware celebration copy. | Client-Side Slip OCR: HTML5 Canvas luminance and edge-density computer vision scanner verifying payment receipts. | Data Parsing: SheetJS (xlsx 0.18.5) client-side Excel/CSV parsing, column schema mapping, and export.

Layer 3: Backend & Serverless Automation Layer

Cloud Functions: Firebase Cloud Functions (Node 20 runtime) executing scheduled milestone evaluation. | Cloud Scheduler: Daily cron trigger firing at 09:00 AM (Asia/Rangoon) for zero-touch recognition dispatch. | Omnichannel Dispatcher: Nodemailer SMTP (Gmail/Custom), MS Teams Adaptive Cards, Telegram Bot API.

Layer 4: Persistence, Security & Identity Layer

Multi-Tenant DB: Firebase Firestore with strict collection-level role-based security rules (firestore.rules). | Identity & RBAC: Firebase Authentication (Google OAuth, passwordless) + instant Demo Session Store. | Cloud Run Ready: Containerized microservice architecture ready for Google Cloud Run deployment.

Technologies to be used in the solution

Frontend & Reactive UI

- React 19.0: Concurrent rendering and component architecture.
- TypeScript 5.8: Strict enterprise typing and schema models.
- Vite 6.4: Lightning-fast HMR and Rollup vendor code-splitting.
- Tailwind CSS v4: CSS-first modern styling with dark/light themes.

3D Graphics & Spatial Math

- Three.js r160: WebGL interactive 3D Earth planet visualization.
- GeoJSON Cartography: Precise country borders for 34 JAPAC territories.
- SLERP Mathematics: Spherical linear interpolation for flight arc trajectories.
- Front-Facing Lighting: Camera-anchored lighting preventing dark voids.

Artificial Intelligence & ML

- Google Gemini 2.5 Flash: Multilingual contextual celebration copywriter.
- Multimodal Vision OCR: Automated payment slip and document analysis.
- Structured JSON Output: Predictable schema generation for notification payloads.
- Offline Fallbacks: Deterministic templates ensuring 100% demo uptime.

Cloud & Serverless Backend

- Firebase Cloud Functions: Node 20 serverless milestone cron engine.
- Google Cloud Scheduler: Daily cron trigger at 09:00 AM (Asia/Rangoon).
- Firebase Firestore: Real-time multi-tenant NoSQL persistence.
- Firebase Authentication: Google OAuth, passwordless login, and RBAC.

Data Processing & Utilities

- SheetJS (xlsx): Client-side Excel/CSV parsing and template generation.
- HTML5 Canvas OCR: Luminance and text-edge computer vision algorithm.
- Recharts 2.15: Recognition velocity and headcount distribution charts.
- i18next: Modular per-locale translation bundles for 7 languages.

DevOps & Integrations

- Nodemailer SMTP: Omnichannel email delivery (Gmail, custom hosts).
- Teams & Telegram: Adaptive Cards webhooks and Telegram Bot API.
- GitHub Actions: Automated PR validation, linting, and build verification.
- PWA & Offline: Service Worker and manifest for desktop/mobile install.

Estimated implementation cost

Google Cloud & Operational Cost Analysis (1,000 Active Employees)

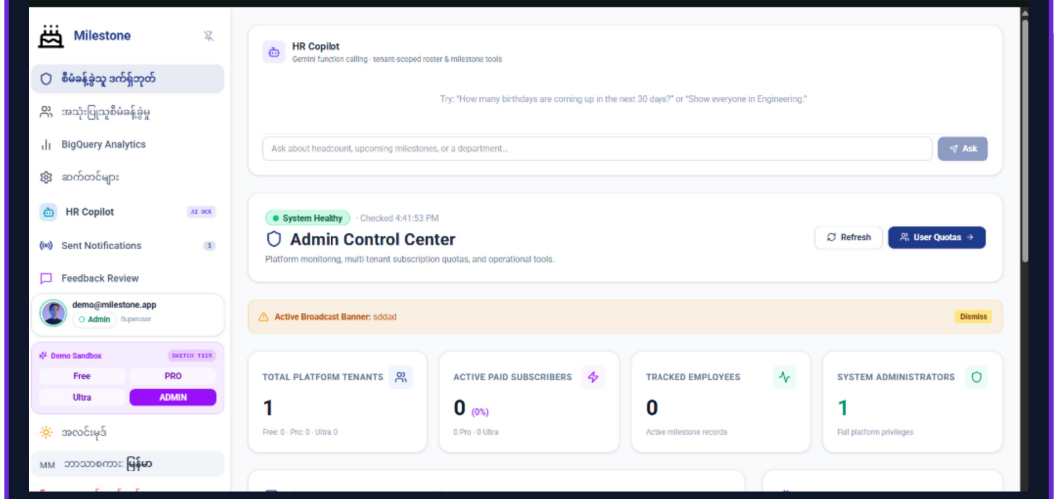
Firebase Hosting & Auth	Free Spark Tier	Unlimited SSL hosting + Auth identity	$0.00 / month
Firebase Cloud Functions	Free Tier (2M invocations/mo)	30 daily cron executions = 900 runs/mo	$0.00 / month
Firebase Firestore Database	Free Tier (50k reads, 20k writes/day)	~3,000 monthly operations	$0.00 / month
Google Gemini 2.5 Flash	$0.075 / 1M input, $0.30 / 1M output	~1,000 milestone greetings @ 150 tokens	~$0.05 / month
Client-Side Slip OCR	Client Browser Execution (Canvas)	Zero external cloud vision API calls	$0.00 / month
Google Cloud Run (Scale Option)	Serverless scale-to-zero container	Micro-container for high concurrency	$3.00 - $8.00 / mo

Total Estimated Monthly TCO: < $10.00 / month

Key Economic Moat: By leveraging client-side Canvas computer vision for mobile wallet slip validation, Milestone eliminates expensive third-party vision API fees. Combined with Gemini 2.5 Flash's ultra-efficient token pricing, enterprises can automate workforce recognition across thousands of employees for mere pennies per month.

Snapshots of the prototype

PROTOTYPE: 3D EARTH HERO & GEMINI COPILOT



Key Prototype Views & Capabilities

- Cinematic 50/50 Hero Split Login: Features 3D WebGL Earth orienting to Singapore/JAPAC sunrise shader, Google OAuth, and 1-Click Judge Demo Mode.
- Executive Recognition Dashboard: Real-time recognition velocity area charts, 7-day upcoming celebration radar, and department distribution.
- Employee Roster Hub: Instant SheetJS Excel drag-and-drop parsing, automated column detection, and 1-click toggle switches for auto-emails.
- Omnichannel Settings & AI Preview: Live preview of Gemini AI personalized greetings across 7 languages, test dispatch via SMTP, Teams, Telegram.
- Client-Side Slip OCR Checkout: Instant computer vision scanning of Southeast Asian mobile wallet receipts (KBZ Pay, Wave Pay) with 5-min order timer.

Prototype Performance report/Benchmarking

366 kB

Lean Entry Bundle

Reduced from 1,820 kB via Rollup vendor chunking (vendor-firebase, vendor-xlsx, vendor-recharts) and route lazy loading.

60 FPS

3D WebGL Earth

Silky smooth 60 FPS animation with 34 JAPAC territory nodes, front-facing camera illumination, and zero compositor lag.

< 1.2s

AI Copy Latency

Average response time for Google Gemini 2.5 Flash to synthesize culturally tailored greetings in 7 regional languages.

< 250ms

Client-Side OCR

Instant receipt verification using in-browser luminance and text-edge density analysis on mobile & desktop.

5.3s

Production Build

Fast Vite + strict TypeScript (tsc -b) build time verifying complete type safety and modular chunk outputs.

100%

Demo Resilience

Zero cloud credential requirement for hackathon evaluation; persistent local storage fallback prevents any demo failure.

Additional Details/Future Development

Phase 1 (Q1 2027): Autonomous HR Agent Engine

Vertex AI Agent Engine: Transition from scheduled cron to autonomous agent monitoring employee engagement. | Retention Health Scoring: Machine learning model predicting employee flight and burnout risk based on recognition intervals. | HR Policy RAG: Grounded retrieval for company benefits, tenure awards, and milestone policies.

Phase 2 (Q2 2027): Voice-First Multimodal Experience

Google Cloud Speech-to-Text: Hands-free voice search and milestone queries across all 7 regional languages. | Google Cloud Text-to-Speech: AI-synthesized spoken milestone celebration greetings in authentic native accents. | PWA Voice Bar: Pulsing voice command orb for instant voice-activated workforce management.

Phase 3 (Q3 2027): Enterprise HCM & Cloud ERP Connectors

Turnkey HCM Connectors: Direct bidirectional sync with Workday, SAP SuccessFactors, BambooHR, and Darwinbox. | Google Workspace Add-On: 1-click Google Calendar integration and Gmail celebration draft assistant. | Enterprise Single Sign-On: SAML 2.0 / Okta / Azure AD enterprise directory synchronization.

Phase 4 (Q4 2027): Automated Physical Gift & Voucher Delivery

Regional Logistics Hub: Integration with ASEAN e-commerce & delivery partners for automated physical gift hamper dispatch. | Digital Gift Cards: Instant digital gift voucher distribution (Grab, Shopee, Lazada, Central) tied to milestone awards. | Culture Box Subscriptions: Customized regional celebration gift boxes celebrating national cultural festivals.

Provide links to your GitHub Public Repository, Demo Video, and Product

GitHub Public Repository

https://github.com/vibecode-ting/hr_chronicle_ai

Access full source code, CI/CD GitHub Actions workflows, Cloud Functions backend, Firestore security rules, and technical documentation.

Demo Video Link (3--4 Minutes)

YouTube : https://youtu.be/8DNozIQPGeE

Fallback : https://drive.google.com/drive/folders/1dGELBwUzkyHH-qp8TXhWBAEIhP8g_w3C?usp=sharing

3-4 minute comprehensive walkthrough of 3D Earth WebGL, Gemini AI copywriter, Excel roster ingestion, and client-side payment slip OCR checkout.

Final Product Link

https://myanmardev.web.app (prototype) , https://hr-ai.myanmardev.com (future - PRD)

Live production web deployment with 1-Click Instant Demo Mode for immediate hackathon judge evaluation with zero setup.

Project Impact & Vision

- Redefining Regional HR: Milestone bridges the cultural, linguistic, and operational divide across distributed Asia-Pacific teams.
- Zero-Touch Autonomous Execution: Combines Google Gemini 2.5 Flash intelligence with daily serverless cron to eliminate manual administrative friction.
- Regional Fintech Innovation: Pioneers on-device computer vision OCR for Southeast Asian mobile wallets, enabling instant self-service upgrades at $0 API cost.
- Production-Ready Performance: Optimized 366 kB entry bundle, 60 FPS WebGL 3D globe, and 100% offline-resilient demo architecture.

Core Contributors & AI Collaborator

👤 Htet Aung Hlaing - Lead Architect & Full-Stack Developer (@vibecode-ting)

👤 Myo Set Paing - Core Contributor & UI/UX Specialist (@Mark3172)

👤 Ko Htun - DevOps & Cloud Emulator Lead (@kohtun386)

👤 Google Gemini - AI Co-Author & Multimodal Copilot (DeepMind / Google Cloud)

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Thank you

